

A dichotomy for the Golden-Parachute boundary in the impatient-agent regime

[Author name]

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Abstract

Possamaï and Touzi show that, in the second-best continuous-time principal-agent problem with face-lifting, the stopping set has the form

$$S = \{0\} \cup [y_{\text{gp}}, \infty)$$

under their regularity and concavity assumptions. They also observe that, when the agent is more impatient than the principal, namely when $\delta > 1$, it is unclear in general whether the upper boundary y_{gp} is finite. This note gives a conditional resolution of that question within the classical Hamilton-Jacobi-Bellman regime used in their verification theorem. The first result is local: no finite positive boundary can occur. The second result identifies the remaining finite case. If Φ denotes the face-lifted retirement reward and $\mathcal{J}$ the effort Hamiltonian in the HJB equation, then

$$y_{\text{gp}} < \infty \iff y_{\text{gp}} = 0 \iff \mathcal{J}(\Phi'(y), \Phi''(y)) \leq 0 \text{ for all } y > 0.$$

Thus, in the impatient-agent regime, finiteness of the Golden-Parachute boundary is equivalent to immediate retirement everywhere. If the value function separates from the face-lifted retirement reward at any positive state, then it never meets it again at a finite positive state.

1 Introduction

The continuous-time contracting model of Possamaï and Touzi [1] extends Sannikov's principal-agent problem [2] by allowing the principal and the agent to have different discount rates. A central object in the second-best problem is the face-lifted retirement reward. Under Assumption 3.3 of [1], their Theorem 3.4 proves that the stopping set has the one-sided form

$$S = \{V = \Phi\} = \{0\} \cup [y_{\text{gp}}, \infty) \tag{1}$$

for some extended threshold $y_{\text{gp}} \in [0, \infty]$, with the convention that $[\infty, \infty)$ is empty. Their Remark 3.5(iii) observes that, in the regime $\delta > 1$, it is unclear in general whether y_{gp} is finite.

This note isolates a simple sign obstruction at a hypothetical finite positive boundary. The obstruction comes from differentiating the first-order equation solved by the face-lifted reward. The sign changes at $\delta = 1$. For $\delta > 1$, it prevents a positive continuation premium $V - \Phi$ from decaying to zero at a finite right endpoint. Consequently,

$$y_{\text{gp}} \in \{0, \infty\}.$$

The remaining question is whether the finite alternative $y_{\text{gp}} = 0$ occurs. This is decided by the sign of the effort Hamiltonian evaluated on the face-lifted reward:

$$y_{\text{gp}} = 0 \iff \mathcal{J}(\Phi', \Phi'') \leq 0 \text{ on } (0, \infty).$$

If this inequality fails at even one positive state, then the no-positive-finite-boundary result forces $y_{\text{gp}} = \infty$.

The proof is conditional on the HJB characterization and regularity conclusions of [1]. In particular, we use that V is C^1 , is C^2 except possibly at one point, and is the unique viscosity solution in the relevant comparison class. The boundary argument itself is classical on an interval avoiding the possible non-classical point. The small technical point made explicit below is that the concave transform f^* is differentiable along the curve $\delta\Phi'$, even if this differentiability is not assumed globally.

2 Equations and standing hypotheses

We use notation close to [1], but denote the face-lifted retirement reward by Φ . Let

$$f := -u^{-1}, \quad f^*(p) := \inf_{x \geq 0} \{xp - f(x)\}, \quad p \leq 0,$$

be the concave Fenchel transform of the original retirement payoff. In the non-degenerate face-lifted case $\delta\gamma > 1$, the function Φ is decreasing and concave, and it satisfies the first-order face-lift equation

$$\Phi(y) - \delta y \Phi'(y) + f^*(\delta \Phi'(y)) = 0, \quad y > 0. \quad (2)$$

We also use the continuous extension $\Phi(0) = 0$, so that Φ has the same boundary value as the value function.

For a twice differentiable function w , define the first-order part of the HJB operator by

$$\mathcal{A}w(y) := w(y) - \delta y w'(y) + f^*(\delta w'(y)), \quad (3)$$

and the full operator by

$$\mathcal{L}w(y) := \mathcal{A}w(y) - \mathcal{J}(w'(y), w''(y))^+, \quad r^+ := \max\{r, 0\}. \quad (4)$$

Here $\mathcal{J}$ is the effort Hamiltonian from [1]. Its precise formula is not needed for the local boundary argument; only the non-negativity of the positive part is used there.

The value function V is characterized by the obstacle problem

$$V(0) = 0, \quad \min\{V - \Phi, \mathcal{L}V\} = 0, \quad (5)$$

in the viscosity sense, and classically at points where V is C^2 . Therefore, on the continuation region $\{V > \Phi\}$, at classical points,

$$\mathcal{A}V(y) = \mathcal{J}(V'(y), V''(y))^+ \geq 0. \quad (6)$$

Equation (2) says simply that

$$\mathcal{A}\Phi(y) = 0, \quad y > 0. \quad (7)$$

Throughout the paper we assume the following classical form of the Possamaï-Touzi regularity conclusions: $V \in C^1(0, \infty)$, V is C^2 away from at most one point, $\Phi \in C^2(0, \infty)$, $\Phi' < 0$, and Φ is concave. We also use the comparison/uniqueness statement for (5) in the class considered in [1].

3 Differentiability along the face-lift curve

The main proof uses the supporting-line inequality for the concave function $f^\star$ at the point $\delta\Phi'(y)$. The following lemma justifies this even if $f^\star$ is not assumed to be differentiable everywhere.

Lemma 1 (Differentiability along the face-lift curve). *Let $J \subset (0, \infty)$ be an interval. Assume $\delta > 1$, $\Phi \in C^2(J)$, $\Phi' < 0$ on J , $\Phi'' \leq 0$ on J , and that g is a finite concave function on an open interval containing*

$$p(J), \quad p(y) := \delta\Phi'(y).$$

Assume also that

$$\Phi(y) - \delta y\Phi'(y) + g(\delta\Phi'(y)) = 0, \quad y \in J. \quad (8)$$

Then, for every $y \in J$,

$$\Phi''(y) < 0,$$

g is differentiable at $\delta\Phi'(y)$, and

$$\delta(g'(\delta\Phi'(y)) - y) = \frac{(\delta - 1)\Phi'(y)}{\Phi''(y)} > 0. \quad (9)$$

Proof. Set

$$r(y) := \delta y\Phi'(y) - \Phi(y), \quad p(y) := \delta\Phi'(y).$$

Then (8) is the identity

$$g(p(y)) = r(y), \quad y \in J. \quad (10)$$

Fix $y_0 \in J$. Since g is finite and concave on an open interval containing $p(y_0)$, it is locally Lipschitz near $p(y_0)$.

We first show that $p'(y_0) \neq 0$. If $p'(y_0) = 0$, then

$$p(y) - p(y_0) = o(|y - y_0|).$$

The local Lipschitz property of g would imply

$$g(p(y)) - g(p(y_0)) = o(|y - y_0|),$$

so $(g \circ p)'(y_0) = 0$. But the right-hand side of (10) is r , and

$$r'(y_0) = (\delta - 1)\Phi'(y_0) + \delta y_0\Phi''(y_0).$$

The condition $p'(y_0) = \delta\Phi''(y_0) = 0$ would therefore give

$$r'(y_0) = (\delta - 1)\Phi'(y_0) \neq 0,$$

because $\delta > 1$ and $\Phi'(y_0) < 0$. This contradiction proves $p'(y_0) \neq 0$. Since $\Phi'' \leq 0$, we have $p'(y_0) = \delta\Phi''(y_0) < 0$, and hence $\Phi''(y_0) < 0$.

By the inverse function theorem, after shrinking to a neighborhood of y_0 , the map p is a C^1 diffeomorphism onto an open interval containing $p(y_0)$. On that open interval,

$$g = r \circ p^{-1}$$

by (10). Hence g is differentiable at $p(y_0)$, and

$$g'(p(y_0)) = \frac{r'(y_0)}{p'(y_0)} = \frac{(\delta - 1)\Phi'(y_0) + \delta y_0\Phi''(y_0)}{\delta\Phi''(y_0)} = y_0 + \frac{(\delta - 1)\Phi'(y_0)}{\delta\Phi''(y_0)}.$$

Multiplying by δ and subtracting δy_0 gives (9). The final positivity follows from $\delta > 1$, $\Phi' < 0$, and $\Phi'' < 0$. $\square$

4 No positive finite boundary

Theorem 1 (No positive finite Golden-Parachute boundary). *Assume $\delta > 1$ and $\delta\gamma > 1$. Work under the standing hypotheses above, and suppose that the stopping set has the form*

$$S = \{y \geq 0 : V(y) = \Phi(y)\} = \{0\} \cup [y_{\text{gp}}, \infty)$$

for some $y_{\text{gp}} \in [0, \infty]$. Then

$$y_{\text{gp}} \in \{0, \infty\}.$$

Equivalently, no finite positive upper stopping boundary can occur.

Proof. Assume for contradiction that

$$0 < y_* := y_{\text{gp}} < \infty.$$

Then

$$V(y) > \Phi(y) \quad \text{for } 0 < y < y_*, \quad V(y) = \Phi(y) \quad \text{for } y \geq y_*. \quad (11)$$

Let

$$D(y) := V(y) - \Phi(y).$$

Thus $D > 0$ on $(0, y_*)$, while continuity gives

$$\lim_{y \uparrow y_*} D(y) = D(y_*) = 0. \quad (12)$$

Apply Lemma 1 with $g = f^*$ on a neighborhood of y_* . The function

$$B(y) := \delta((f^*)'(\delta\Phi'(y)) - y) \quad (13)$$

extends continuously to y_* by the identity

$$B(y) = \frac{(\delta - 1)\Phi'(y)}{\Phi''(y)}. \quad (14)$$

Moreover $B(y_*) > 0$. Hence there are $a_0 < y_*$ and $b > 0$ such that

$$B(y) \geq b, \quad a_0 < y \leq y_*. \quad (15)$$

Since V is C^2 except possibly at one point, we may choose $a \in (a_0, y_*)$ so that V is C^2 on

$$I := (a, y_*).$$

On I , the value function is in the continuation region. By (6) and (7),

$$\mathcal{A}V(y) \geq 0, \quad \mathcal{A}\Phi(y) = 0, \quad y \in I.$$

Subtracting these two relations yields

$$0 \leq D(y) - \delta y D'(y) + f^*(\delta V'(y)) - f^*(\delta \Phi'(y)). \quad (16)$$

For a concave function differentiable at q , the supporting-line inequality gives

$$f^*(p) - f^*(q) \leq (f^*)'(q)(p - q).$$

Taking $p = \delta V'(y)$ and $q = \delta \Phi'(y)$, and using Lemma 1, we obtain

$$f^*(\delta V'(y)) - f^*(\delta \Phi'(y)) \leq \delta (f^*)'(\delta \Phi'(y)) D'(y). \quad (17)$$

Combining (16) and (17),

$$0 \leq D(y) + B(y)D'(y), \quad y \in I. \quad (18)$$

Since $B > 0$ on I , this is equivalent to

$$D'(y) + \frac{1}{B(y)}D(y) \geq 0, \quad y \in I. \quad (19)$$

Fix $y_0 \in I$, and define

$$M(y) := D(y) \exp \left(\int_{y_0}^y \frac{ds}{B(s)} \right), \quad y \in [y_0, y_*).$$

By (19), $M'(y) \geq 0$. Therefore, for every $y \in [y_0, y_*)$,

$$D(y) \geq D(y_0) \exp \left(- \int_{y_0}^y \frac{ds}{B(s)} \right).$$

Using (15),

$$\int_{y_0}^y \frac{ds}{B(s)} \leq \frac{y_* - y_0}{b}, \quad y_0 \leq y < y_*.$$

Hence

$$D(y) \geq D(y_0) \exp \left(- \frac{y_* - y_0}{b} \right) > 0, \quad y_0 \leq y < y_*.$$

Taking $y \uparrow y_*$ contradicts (12). Thus $0 < y_{\text{gp}} < \infty$ is impossible. $\square$

5 The finite/infinite dichotomy

The preceding result leaves two alternatives: immediate contact everywhere, $y_{\text{gp}} = 0$, or no positive contact at all, $y_{\text{gp}} = \infty$. The following proposition identifies which one occurs.

Proposition 1 (Classification of the degenerate finite case). *Under the hypotheses of Theorem 1, define*

$$Q(y) := \mathcal{J}(\Phi'(y), \Phi''(y)), \quad y > 0.$$

Then

$$y_{\text{gp}} < \infty \iff y_{\text{gp}} = 0 \iff Q(y) \leq 0 \text{ for every } y > 0. \quad (20)$$

Equivalently, exactly one of the following two alternatives holds:

- (i) if $Q \leq 0$ on $(0, \infty)$, then $V = \Phi$ on $[0, \infty)$ and $y_{\text{gp}} = 0$;
- (ii) if $Q(y) > 0$ for some $y > 0$, then $V > \Phi$ on $(0, \infty)$ and $y_{\text{gp}} = \infty$.

Proof. The face-lift equation (7) gives

$$\mathcal{L}\Phi(y) = \mathcal{A}\Phi(y) - \mathcal{J}(\Phi'(y), \Phi''(y))^+ = -Q(y)^+. \quad (21)$$

Assume first that $Q \leq 0$ on $(0, \infty)$. Then $Q^+ = 0$, and (21) gives $\mathcal{L}\Phi = 0$. Since also $\Phi - \Phi = 0$, the function Φ satisfies

$$\min\{\Phi - \Phi, \mathcal{L}\Phi\} = 0$$

classically, with the same boundary and admissibility conditions as the face-lifted payoff. By the uniqueness/comparison theorem for the variational inequality (5), $V = \Phi$. Therefore $S = [0, \infty)$ and $y_{\text{gp}} = 0$.

Conversely, assume $y_{\text{gp}} = 0$. Then (1) gives $V = \Phi$ on $[0, \infty)$. Inserting this equality into the variational inequality at classical points yields

$$0 = \min\{0, \mathcal{L}\Phi(y)\}, \quad y > 0.$$

Thus $\mathcal{L}\Phi(y) \geq 0$. But (21) says $\mathcal{L}\Phi(y) = -Q(y)^+ \leq 0$. Hence $Q(y)^+ = 0$, or equivalently $Q(y) \leq 0$, for every $y > 0$.

Finally, Theorem 1 says that y_{gp} is either 0 or ∞ . Therefore, if $Q(y) > 0$ somewhere, the equivalence just proved rules out $y_{\text{gp}} = 0$, and the only remaining possibility is $y_{\text{gp}} = \infty$. This proves (20) and the two alternatives. $\square$

Corollary 1 (Answer to the finiteness question in the classical regime). *In the regime $\delta > 1$, under the Possamai-Touzi hypotheses and the classical regularity/uniqueness assumptions stated above, the upper boundary is finite if and only if the face-lifted reward itself solves the HJB variational inequality. Equivalently,*

$$y_{\text{gp}} < \infty \iff \mathcal{J}(\Phi', \Phi'') \leq 0 \quad \text{on } (0, \infty).$$

If this condition fails, then $y_{\text{gp}} = \infty$.

6 Comments on the sign mechanism

The no-positive-finite-boundary theorem is entirely local at a hypothetical finite boundary. It uses only two features of the equations. First, in the continuation region the full HJB equation implies that the first-order part is non-negative:

$$\mathcal{A}V = \mathcal{J}(V', V'')^+ \geq 0.$$

Second, the face-lifted reward satisfies the exact first-order equation $\mathcal{A}\Phi = 0$. Subtracting these relations and using concavity of f^* gives the one-dimensional inequality

$$D + BD' \geq 0, \quad D := V - \Phi.$$

The face-lift equation identifies the coefficient as

$$B = \delta ((f^*)'(\delta\Phi') - y) = \frac{(\delta - 1)\Phi'}{\Phi''}.$$

When $\delta > 1$, the signs $\Phi' < 0$ and $\Phi'' < 0$ imply $B > 0$. A positive function satisfying $D + BD' \geq 0$, with B bounded away from zero near the endpoint, cannot hit zero at a finite right endpoint. This is the obstruction to a finite positive Golden-Parachute boundary.

At $\delta = 1$, the coefficient degenerates. For $\delta < 1$, its sign is reversed. Thus the argument is specific to the impatient-agent regime $\delta > 1$, which is the regime in which Remark 3.5(iii) of [1] leaves the finiteness of y_{gp} unresolved.

Remark 1 (On viscosity formulations). The proof above is written at the classical level because it uses pointwise differentiation of the face-lift equation and the pointwise HJB equation on an interval where V is C^2 . This is sufficient under the regularity theorem quoted from [1]. If one works only with the viscosity characterization and does not import this regularity, the same sign obstruction should be implemented with jets or by a local smoothing/regularization argument. That purely viscosity proof is not attempted here.

Remark 2 (What is, and is not, proved). The result does not claim that y_{gp} is always infinite when $\delta > 1$. Rather, it says that finiteness is equivalent to the degenerate immediate-retirement case $V = \Phi$. The condition $\mathcal{J}(\Phi', \Phi'') \leq 0$ is exactly the condition under which the face-lifted reward already satisfies the full HJB variational inequality.

References

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